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Survival machine learning methods for mortality prediction after heart transplantation in the contemporary era

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Abstract

Although prediction models for heart transplantation outcomes have been developed previously, a comprehensive benchmarking of survival machine learning methods for mortality prognosis in the most contemporary era of heart transplants following the 2018 donor heart allocation policy change is warranted. This study assessed seven statistical and machine learning algorithms–Lasso, Ridge, Elastic Net, Cox Gradient Boost, Extreme Gradient Boost Linear, Extreme Gradient Boost Tree, and Random Survival Forests in a post-policy cohort of 7,160 adult heart-only transplant recipients in the Scientific Registry of Transplant Recipients (SRTR) database who received their first transplant on or after October 18, 2018. A cross-validation framework was designed in *mlr*. Model performance was also compared in a seasonally-matched pre-policy cohort. In the post-policy cohort, Random Survival Forests and Cox Gradient Boost had the highest performances with C-indices of 0.628 and 0.627. The relative importance of some predictive variables differed between the pre-policy and post-policy cohorts, such as the absence of ECMO in the post-policy cohort. Survival machine learning models provide reasonable prediction of 1-year posttransplant mortality outcomes and continual updating of prediction models is warranted in the contemporary era.

Introduction

Orthotopic heart transplantation (HTx) remains the gold standard therapy for end-stage heart failure. Careful patient selection has allowed for a contemporary median survival of 12.5 years

and Transplantation Network. The authors had permission to use the SRTR dataset under project number #9989. The data are available from SRTR upon request by emailing srtr@srtr.org (More information can be found at https://www.srtr.org/requesting-srtr-data/data-requests/). SRTR is a third-party organization that owns the data, thus the authors are not allowed to share data per the Data Use Agreement that was signed with SRTR. The authors confirm they did not have any special privileges that other authors would not have.

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Abbreviations: BMI, body mass index; CPRA, calculated panel reactive antibody; ECMO, extracorporeal membrane oxygenation; GFR, glomerular filtration rate; IABP, intra-aortic blood pump; ICU, intensive care unit; LOS, length of stay; OPTN, Organ Procurement and Transplantation Network; PCW, pulmonary capillary wedge pressure; PVR, pulmonary vascular resistance; SRTR, Scientific Registry of Transplant Recipients; VAD, ventricular assist device.

with an additional 2 years conditional upon 1-year survival post-transplant [1]. Accurate determination of which patient characteristics should be considered contraindications to transplantation for optimal outcomes remains one of the greatest challenges in our field. Historically, heart transplant risk prediction models have been generated using multivariable logistic regression [2] or Cox proportional hazard regression [3,4] and have yielded the identification of certain key risk factors. The limitation of such models lies in the assumption of purely linear relationships that do not account for more complex associations. Although many factors associated with post-transplantation survival have been identified [5,6], there remains variability in models exploring more complex relationships. Recently, several studies have attempted to improve the prediction of heart transplant outcomes using machine learning (ML) methods [7–11] and survival ML methods [12–14] in both adult and pediatric populations. Compared to traditional regression-based modeling approaches, ML algorithms can capture more complex interactions between variables. Accounting for time-to-event information tends to result in more statistically powerful prediction estimates [15]. Furthermore, posttransplantation survival is known to be non-proportional across different strata, making conventional statistical methods such as Cox proportional hazards regression potentially invalid and less powerful for assessing post-transplantation mortality among heart transplant recipients.

Another potentially important consideration is the revision of the donor heart allocation system on October 18, 2018, to "reduce wait-list mortality, enhance geographic organ sharing, and improve organ distribution equity" [16]. Although it has been shown that 1-year post-transplant survival under the new heart allocation policy was not significantly different from before [17], it is possible that both the clinical profile of transplant recipients and clinical care practices have shifted in the almost 5 years since policy implementation [18]. Thus, in this study, we explore and benchmark the performance of machine learning survival algorithms in the post-2018 transplant era (henceforth referred to as "post-policy") to better understand prognostic factors associated with one-year mortality. We compare these results to a parallel analysis run in a seasonally-matched pre-2018 transplant era cohort (henceforth referred to as "pre-policy").

Methods

Data source

We used data from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donor, waitlisted candidates, and transplant recipients in the US, submitted by the Organ Procurement and Transplantation Network (OPTN) members. The data is anonymized such that it is not possible to identify individual participants. The Health Resources and Services Administration (HRSA), U.S. Department of Health and Human Services, provides oversight of the activities of the OPTN and SRTR contractors. We had approval from SRTR under DUA #9899 to conduct this research study. Informed consent is obtained by OPTN for living donors. As formally declared by the UNOS Ethics Committee, no organs were procured from prisoners. We confirm that all research was performed in accordance with relevant guidelines/regulations. The data was last accessed on May 5, 2023. This study was reviewed and informed consent was waived by the Institutional Review Board at Harvard T.H. Chan School of Public Health due to the retrospective and anonymized nature of this registry dataset.

Study population

We excluded recipients <18 years of age at transplant (n = 1,275) and an additional 8,187 recipients of multi-organ transplants because allocation criteria are different for multi-organ

and pediatric candidates. This study includes 7,160 adult heart-only transplant recipients who received their first transplant on or after October 18, 2018, with at least one recorded follow-up visit. We had follow-up data up until June 3, 2021. This study was reviewed and approved by the Institutional Review Board at the Harvard T.H. Chan School of Public Health.

Data preparation

The outcome of interest was one-year all-cause mortality assessed from the time of heart transplantation to death or end of the one-year follow-up. Survival data for recipients in both cohorts were administratively censored at 1 year after transplant to prevent bias from the differential length of follow-up between cohorts. For each cohort, we split 90% as a training set and 10% as a holdout set. In order of operation, we performed z-score standardization on continuous variables, multiple imputation, and one-hot encoding on categorical variables separately in the validation set and holdout set to minimize data leakage. We included predictor variables based on expert opinion and literature review. We excluded potential predictors with over 20% missingness. For those with less than 20% missingness, we used multiple imputation with chained equations with 5 iterations using the R package *mice* [19], which preserves the associations in the data and the uncertainty in those associations. We assumed that the data are missing at random and that censoring is non-informative. We used one-hot encoding to convert categorical variables into binary variables for each level of each categorical feature. Certain variables that had many levels with only a small number of observations were combined (Supplementary Appendix). This data pre-processing led to a final set of 75 demographic, clinical, recipient, waitlist, donor, and procedural variables (114 after one-hot encoding), of which 9 were categorical, 44 binary, and the remaining 22 numeric (S1 Table in S1 File).

We also prepared a seasonally matched pre-policy cohort of patients transplants from November 1, 2014, with follow-up time to June 3, 2017, as a sensitivity analysis to account for known seasonal trends in decreased donor heart donation [20]. This cohort ends 1 year before policy implementation to avoid bias from any anticipatory practice changes before the policy implementation in October 2018.

Machine learning analysis

We compared a Cox proportional hazards model and 7 machine learning algorithms. Lasso, Elastic Net, and Ridge regression were selected as the penalized Cox regression methods [21,22]. Boosted Cox regression methods included a Cox model with gradient boosting (Cox Boost) [23], Extreme Gradient Boosting (XGBoost) [24] with linear model-based boosting (XGBL), and tree-based boosting XGBoost (XGBT) [25]. Random Survival Forests (RSF) [26] was chosen as the tree-based ensemble method. Details on the models and selected parameters are described in the Supplementary Appendix. Each model was trained using a nested 5-repeat, 5-fold cross-validation within the validation cohort (80/20 split), and tuning was automatically performed for the machine learning algorithms to select optimal hyperparameters. A random search with 25 iterations was used to select values for the hyperparameters in the inner loop and model performance was evaluated in the outer loop. For each method, we took the parameters from the best-performing model within the cross-validation and used the entirety of the training cohort to produce the final model. The final model was then tested on the holdout set. The R package *mlr* (Machine Learning in R) [27] was used as a framework to carry out benchmarking of these experiments. A graphical schematic of our benchmark framework is presented in Fig 1. To evaluate model performance, we calculated both the cross-validated and holdout concordance index (Harrell's C-index). The C-index measures the

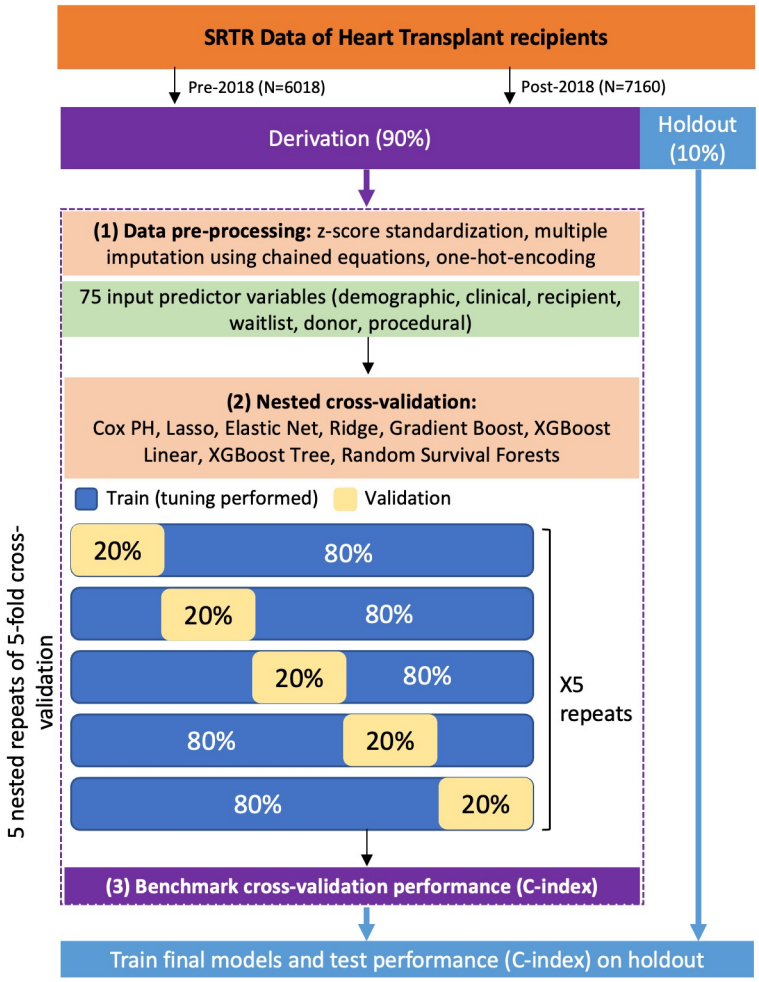

Fig 1. Schematic of machine learning benchmarking study design.

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proportion of pairs where the observation with the higher survival time has the higher probability of survival as predicted by the model. A C-index of 1 means the predictions were perfect: higher-risk patients are ranked ahead of lower-risk patients. A C-index of 0.5 means the predictions are random: half the time higher-risk patients are correctly ranked ahead, half the time lower-risk patients are incorrectly ranked ahead. Variable weights were computed as the beta coefficients from regression-based models and importance from boosted and ensemble models. Variable relevance was determined as the number of times it was selected across 25 cross-validation iterations. Variable significance was defined as either the empirical cross-validated 95% confidence interval of the hazard ratio not crossing 1 or the importance weight not crossing 0. To assess whether there were statistical differences between cross-validated models, we calculated the corrected resampled paired t-test, proposed by Nadeau and Bengio [28], using the R package *correctR*. A standard t-test on results from a repeated k-fold cross-validation inflates the Type I error is inflated because the violation of the independence assumption leads to an underestimation in the variance. A Bonferroni adjustment was applied to p-values to control the false discovery rate at 0.05. All analyses were performed using R 4.3.0. The code used in the analysis can be found at https://github.com/latlio/srtr_mlr.

Results

Cohort characteristics

In the post-policy cohort, 506 patients died within one year of their heart transplant during the study period from October 18, 2018, to June 3, 2021. The crude one-year survival was 90.9% (95% CI: 90.2%, 91.7%) with 5864 total person-years of follow-up time. The average recipient age was 53.5, and the average donor age was 32.5. A total of 72.5% (n = 5193) of the cohort was male, and 72.1% (n = 5162) of the cohort was white. 27.1% had diabetes. 37.6% of patients were bridged with ventricular assist device (VAD), 4.9% of patients were bridged with extra-corporeal membrane oxygenation (ECMO), and 26.8% of patients were bridged with intra-aortic balloon pump therapy (IABP). Patients spent a median of 33 days on the wait-list and received organs that spent 3.4 hours of ischemia time on average during transport. A comprehensive summary of demographic, clinical, recipient, waitlist, donor, and procedural variables for the post-policy cohort is detailed in Table 1. We note that in the pre-policy cohort, 512 patients died within one year, and the crude one-year survival rate was 91.5% (95% CI: 90.8%, 92.2%).

Table 1. Baseline characteristics of heart transplant recipients from October 18, 2018 to June 3, 2021.

	Overall (N = 7160)
Allocation Status	
Previous 3-tier system	27 (0.4%)
2018 Status 1	587 (8.2%)
2018 Status 2	3335 (46.6%)
2018 Status 3	1396 (19.5%)
2018 Status 4	1427 (19.9%)
2018 Status 5	0 (0%)
2018 Status 6	388 (5.4%)
BMI (kg/m2)	
Mean (SD)	27.8 (5.03)
Median [Min, Max]	27.5 [11.7, 47.5]
Highest Education Received	
Below High School	243 (3.4%)
High School Graduate	2549 (35.6%)
College Graduate and beyond	4036 (56.4%)
Gender	
Male	5193 (72.5%)
Female	1967 (27.5%)
Insurance	
Medicaid	968 (13.5%)
Medicare	2372 (33.1%)
Private	3262 (45.6%)
Other	314 (4.4%)
Age	
Mean (SD)	53.5 (12.9)
Median [Min, Max]	57.0 [18.0, 76.0]
Cardiac Etiology	
Valvular Heart Disease	74 (1.0%)
Dilated Myopathy	1302 (18.2%)
Ischemic Dilated Myopathy	1900 (26.5%)

(*Continued*)

Table 1. (Continued)

	Overall (N = 7160)
Restrictive Myopathy	330 (4.6%)
Coronary Artery Disease	132 (1.8%)
Other	653 (9.1%)
Idiopathic	2701 (37.7%)
Dialysis	
No	7049 (98.4%)
Yes	111 (1.6%)
Type 2 Diabetes	
No	5174 (72.3%)
Yes	1938 (27.1%)
Previous Malignancy	
No	6530 (91.2%)
Yes	630 (8.8%)
Prior Cardiac Surgery	
No	5644 (78.8%)
Yes	1202 (16.8%)
Steroid Use	
No	6441 (90.0%)
Yes	353 (4.9%)
Cytomegalovirus	
Negative	3075 (42.9%)
Positive	3689 (51.5%)
Epstein-Barr Virus	
Negative	646 (9.0%)
Positive	5910 (82.5%)
Functional Status At Time of Transplant	
Moribund/hospitalized/severely disabled	4302 (60.1%)
Significant Assistance	1638 (22.9%)
Normal	518 (7.2%)
HBV Antibody	
Negative	6325 (88.3%)
Positive	307 (4.3%)
HBV Surf Antigen	
Negative	6664 (93.1%)
Positive	100 (1.4%)
Hepatitis C Virus	
Negative	6593 (92.1%)
Positive	149 (2.1%)
HIV Status	
No	6733 (94.0%)
Yes	37 (0.5%)
Time on Waitlist	
Mean (SD)	186 (385)
Median [Min, Max]	33.0 [0, 5150]
Blood Type	
A	2852 (39.8%)
AB	370 (5.2%)

(*Continued*)

Table 1. (Continued)

	Overall (N = 7160)
B	1100 (15.4%)
O	2838 (39.6%)
Accept a Donor After Cardiac Death	
No	6720 (93.9%)
Yes	440 (6.1%)
Accept a Hepatitis B Positive Donor	
No	3167 (44.2%)
Yes	3993 (55.8%)
Accept a Hepatitis C Positive Donor	
No	2401 (33.5%)
Yes	4759 (66.5%)
Accept a Donor with History of Heart Disease	
No	2856 (39.9%)
Yes	3218 (44.9%)
Symptomatic Cerebrovascular Disease	
No	6556 (91.6%)
Yes	512 (7.2%)
Candidate Functional Status	
Moribund/hospitalized/severely disabled	3870 (54.1%)
Significant Assistance	2275 (31.8%)
Normal	663 (9.3%)
Candidate Preliminary Cross-Matching Required	
No	6642 (92.8%)
Yes	518 (7.2%)
Waitlist Glomerular Filtration Rate (MDRD)	
Mean (SD)	68.7 (27.8)
Median [Min, Max]	64.2 [4.91, 522]
Glomerular Filtration Rate (MDRD) At Time of Transplant	
Mean (SD)	70.1 (30.0)
Median [Min, Max]	65.1 [6.84, 494]
Number of HLA-A Mismatches	
Mean (SD)	1.51 (0.603)
Median [Min, Max]	2.00 [0, 2.00]
Number of HLA-B Mismatches	
Mean (SD)	1.73 (0.481)
Median [Min, Max]	2.00 [0, 2.00]
Number of HLA-DR Mismatches	
Mean (SD)	1.56 (0.580)
Median [Min, Max]	2.00 [0, 2.00]
Received ECMO Pre-transplant	
No	6810 (95.1%)
Yes	350 (4.9%)
Specific HLA Antibody	
No	5674 (79.2%)
Yes	728 (10.2%)
HLA Typing Done	
No	525 (7.3%)

(*Continued*)

Table 1. (Continued)

	Overall (N = 7160)
Yes	6411 (89.5%)
Received IABP	
No	5238 (73.2%)
Yes	1922 (26.8%)
Received Inotropes	
No	4531 (63.3%)
Yes	2629 (36.7%)
Received Life Support (e.g. ECMO, IABP, PGE, Inotropes, Ventilator, NO, VAD)	
No	1196 (16.7%)
Yes	5663 (79.1%)
Received Other Type of Life Support	
No	6791 (94.8%)
Yes	369 (5.2%)
Medical Condition	
ICU	3559 (49.7%)
Hospitalized Not in ICU	907 (12.7%)
Not Hospitalized	2389 (33.4%)
Number of HLA Mismatches	
Mean (SD)	4.69 (1.01)
Median [Min, Max]	5.00 [0, 6.00]
Total Bilirubin (mg/dL)	
Mean (SD)	1.01 (1.78)
Median [Min, Max]	0.700 [0.100, 51.0]
Received Transfusion(s)	
No	5861 (81.9%)
Yes	950 (13.3%)
Received Ventilator Support	
No	5758 (80.4%)
Yes	1061 (14.8%)
VAD Usage	
No	4232 (59.1%)
Yes	2695 (37.6%)
Donor Blood Type	
A	2557 (35.7%)
AB	134 (1.9%)
B	787 (11.0%)
O	3682 (51.4%)
Donor Age	
Mean (SD)	32.5 (10.6)
Median [Min, Max]	32.0 [9.00, 70.0]
Donor Anti-CMV Antibody	
Negative	2735 (38.2%)
Positive	4392 (61.3%)
Donor Anti-HCV Antibody	
Negative	6345 (88.6%)
Positive	815 (11.4%)
Donor Cause of Death	

(*Continued*)

Table 1. (Continued)

	Overall (N = 7160)
Anoxia	3202 (44.7%)
Cerebrovascular/Stroke	956 (13.4%)
Head Trauma	2795 (39.0%)
CNS Tumor	23 (0.3%)
Donor Serum Creatinine (mg/dL)	
Mean (SD)	1.67 (1.76)
Median [Min, Max]	1.04 [0.0400, 17.9]
DDVAP Administered to Donor	
No	6433 (89.8%)
Yes	552 (7.7%)
Donor Gender	
Male	5132 (71.7%)
Female	2028 (28.3%)
Donor Height (cm)	
Mean (SD)	174 (9.45)
Median [Min, Max]	175 [122, 206]
Donor Serum Creatinine >1.5 mg/dL	
No	5164 (72.1%)
Yes	1994 (27.8%)
Donor History of Cancer	
No	7008 (97.9%)
Yes	66 (0.9%)
Donor Cigarette Use >20 Pack-years	
No	6149 (85.9%)
Yes	855 (11.9%)
Donor History of Cocaine Use	
No	4981 (69.6%)
Yes	1875 (26.2%)
Donor History of Diabetes	
No	278 (3.9%)
Yes	6816 (95.2%)
Donor History of Hypertension	
No	1104 (15.4%)
Yes	5982 (83.5%)
Donor History of Drug Use	
No	2530 (35.3%)
Yes	4372 (61.1%)
Donor History of Inotrope Use	
No	4530 (63.3%)
Yes	2448 (34.2%)
Donor Cardiac Arrest	
No	6487 (90.6%)
Yes	673 (9.4%)
Donor Weight (kg)	
Mean (SD)	84.9 (20.2)
Median [Min, Max]	81.8 [39.9, 225]
Recipient Cardiac Output (L/min)	

(*Continued*)

Table 1. (Continued)

	Overall (N = 7160)
Mean (SD)	4.43 (1.43)
Median [Min, Max]	4.27 [0.200, 14.3]
Total Ischemic Time (min)	
Mean (SD)	206 (64.8)
Median [Min, Max]	205 [20.0, 720]
Pulmonary Capillary Wedge Pressure (mmHg)	
Mean (SD)	17.8 (8.73)
Median [Min, Max]	17.0 [0, 50.0]
Transplant Procedure Type	
Orthotopic Bicaval	5677 (79.3%)
Orthotopic Traditional	977 (13.6%)
Orthotopic Total	188 (2.6%)
Heterotopic	0 (0%)
Arterial Diastolic Blood Pressure (mmHg)	
Mean (SD)	19.3 (8.67)
Median [Min, Max]	18.0 [0, 110]
Arterial Mean Blood Pressure (mmHg)	
Mean (SD)	27.0 (10.1)
Median [Min, Max]	26.0 [0, 110]
Arterial Systolic Blood Pressure (mmHg)	
Mean (SD)	39.6 (13.8)
Median [Min, Max]	38.0 [0, 158]

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Machine learning benchmark performance

In the post-policy holdout data, RSF performed the best with a C-index of 0.628, while Cox performed the worst with a C-index of 0.585 (Table 2). Cox Boost performed similarly well with a C-index of 0.627. Both Lasso and Elastic Net performed moderately well in the holdout with C-indices of 0.613 each, although their performance in the cross-validation was poor with cross-validated mean C-indices of 0.516 and 0.508 respectively (S3 Table in S1 File). The average cross-validated C-index ranged from 0.615 to 0.508 (Fig 2). Although XGBL had the highest cross-validation C-index (mean = 0.615 [SD = 0.025]), it did not have the highest holdout C-index. The most variable algorithms were Lasso (SD = 0.033) and Ridge (SD = 0.031) across

Table 2. C-indices for holdout data.

Model	Pre-Policy Model Tested on Pre-Policy Holdout	Post-Policy Model Tested on Post-Policy Holdout	Pre-Policy Model Tested on Post-Policy Holdout
Coxph	0.610	0.585	0.573
Ridge	0.617	0.586	0.577
Lasso	0.626	0.613	0.616
Elastic Net	0.646	0.613	0.620
Cox Boost	0.654	0.627	0.633
XGBoost Linear	0.646	0.600	0.585
XGBoost Tree	0.650	0.612	0.631
Random Survival Forests	0.656	0.628	0.616

https://doi.org/10.1371/journal.pone.0313600.t002

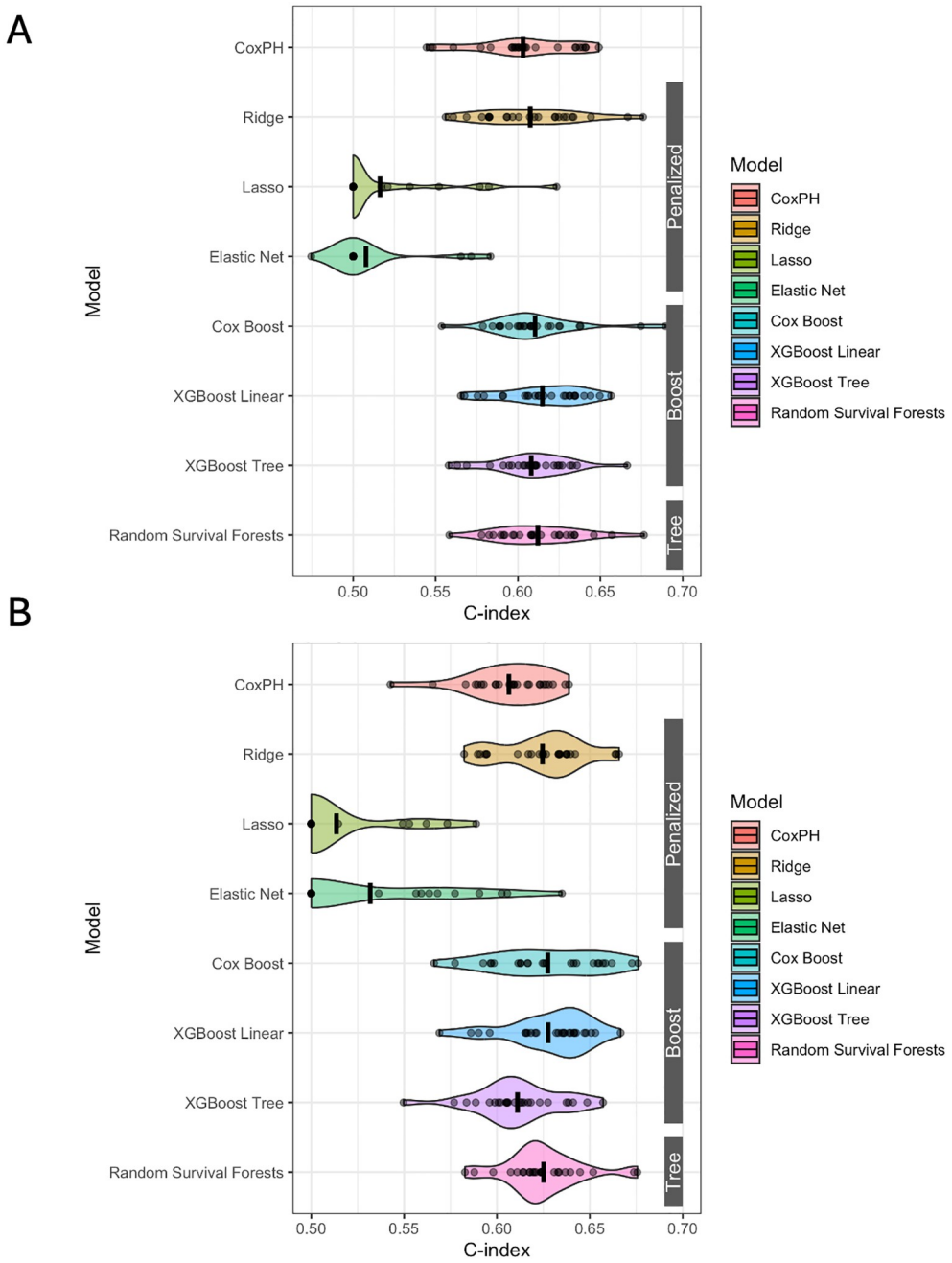

Fig 2. Violin plots comparing cross-validation C-indices for 7 different machine learning survival models against Cox PH in the A) post-policy cohort and B) pre-policy cohort. Black bar indicates the mean c-index.

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cross-validation repeats. Except for Lasso and Elastic Net, there were no statistically significant differences in cross-validation C-indices between Cox and the ML methods (S4 Table in S1 File). Lasso and Elastic Net had significantly worse cross-validated performance than the other ML methods.

In the pre-policy cohort, the algorithms performed similarly, with Cox Boost (holdout C-index = 0.633) performing the best and Cox (holdout C-index = 0.573) the worst (Table 2).

Cross-validation C-indices ranged from 0.628 to 0.514, and the most variable algorithm was Lasso (SD = 0.044) (S4 Table in S1 File). There were also largely no statistically significant differences in cross-validation C-indices between Cox and the ML methods except for Lasso (S4 Table in S1 File). All of the pre-policy cohort-trained models applied to the post-policy holdout set performed more poorly compared to the pre-policy holdout set.

Prognostic variables for posttransplant one-year mortality

Sparse models such as Lasso, Elastic Net, and Cox Boost use a regularization term to shrink coefficients of variables to zero, effectively generating a subset of variables. Hereafter, we refer to this as a model's "selection", although we clarify that we did not perform computational feature selection before training the model. Since we performed 5 repeats of 5-fold cross-validation, a variable can be selected 25 times at most. We use this crude measure of selection to first determine which variables tend to be included in the majority of model replications [29], although we note that there exists inherent variability in sparse variable selection [30]. Age, total ischemic time, bilirubin, BMI, and ECMO were the top 5 most selected variables in the post-policy cohort (Fig 3). The top 5 in the pre-policy cohort were similar except for HIV status. VAD, GFR, prior transfusions, and Medicaid were other variables that were frequently selected in the post- and pre-policy cohorts. Comprehensive heatmap summaries of the total number of times each input predictor was selected in each model in the post- and pre-policy cohorts are provided in S1 and S2 Figs in S1 File respectively.

Plots of the top 20 most important features for the top-performing models are shown in Fig 4. The most important predictor variables are fairly constant between post-policy and pre-policy cohorts, although there are some differences. Some of the most important variables RSF selected in the post-policy cohort were bilirubin, age, pulmonary diastolic pressure, donor height, and wait time (Fig 4A). In the pre-policy cohort, bilirubin, donor age, eGFR measured at waitlist and pre-transplant, and age were selected (Fig 4B). Cox Boost selected HCV positive as the most important variable in the post-policy cohort and HIV negative as the most important variable in the pre-policy cohort (Fig 4C and 4D). XGBT selected bilirubin, eGFR, age, and BMI as the most important variables in both pre and post-policy cohorts (Fig 4E and 4F). Boxplots for all variables hazard ratios and importance scores for all models are shown in S3-9 Figs in S1 File.

Discussion

In this study, we benchmarked survival machine learning methods for the prognosis of one-year all-cause mortality in adult heart transplant recipients transplanted after the UNOS allocation policy change in October 2018 [16]. We also compared the models in a pre-policy era to assess whether the characteristics of predictor variables changed. In both the post- and pre-policy eras, the discriminatory power of machine learning models for one-year all-cause mortality tended to be higher compared to Cox, although not statistically different according to cross-validated C-indices. While the ML models were not statistically superior to Cox, these complex models likely have an advantage over Cox regression in two aspects: they do not require the proportional hazards assumption and they can capture nonlinear interactions between outcome and variables. Thus, these models can potentially provide a level of insight into the different predictive factors in these two eras that linear models cannot. It is also important to state that first, we only used information available pretransplant since that is where a post-transplant mortality prediction model is most likely useful, unlike other prediction models [31]. This may have limited the maximum possible performance of machine learning performance on mortality prediction. Second, based on our study design, we did not use

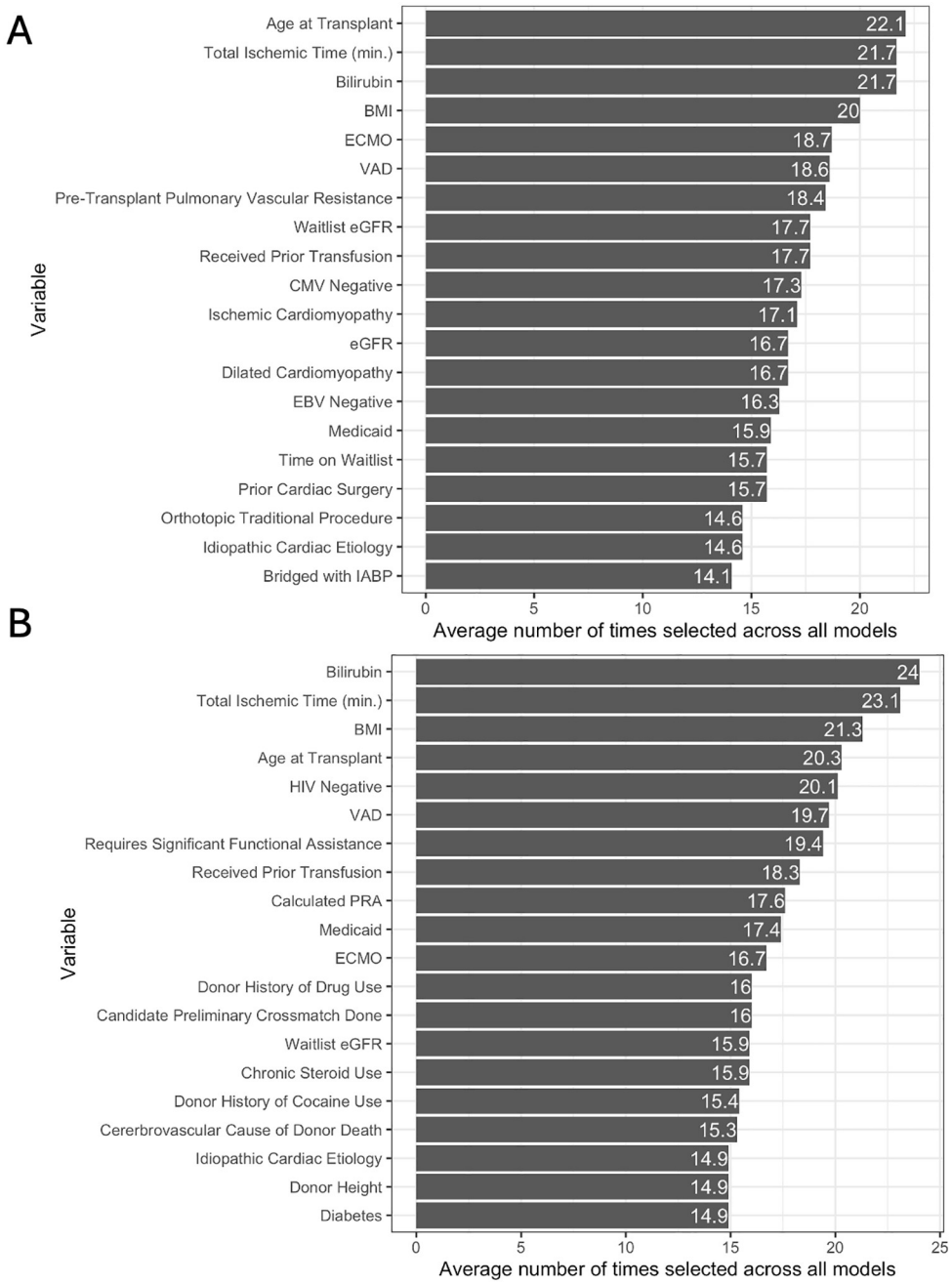

Fig 3. Average number of times variable is selected across all sparse machine learning models in the A) post-policy cohort and B) pre-policy cohort. Sparse machine learning models include Lasso, Elastic Net, and Cox Boost.

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the full range of data available, as other predictive models have done, so our model performance may be affected by lower statistical power. The pre-policy cohort-trained models had lower C-indices when applied to a post-policy holdout set compared to a pre-policy holdout set. This could be explained by underlying changes in treatment practices following the policy change in 2018 such as increased IABP usage and increased pre-transplant temporary mechanical circulatory support use [17,18,32–34].

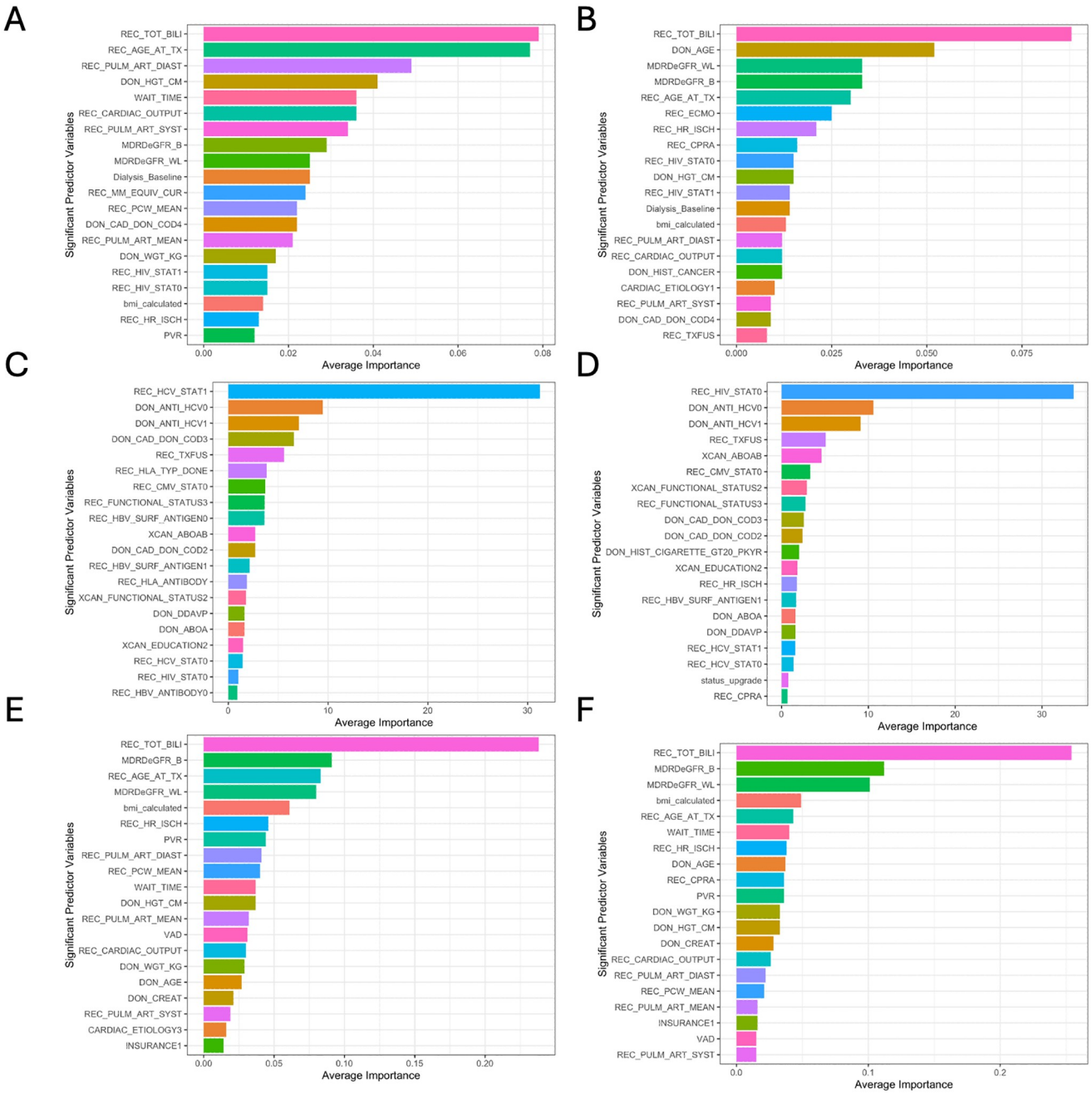

Fig 4. Top 20 significant predictor variables. A) RSF in post-policy cohort B) RSF in pre-policy cohort C) Cox Boost in post-policy cohort D) Cox Boost in pre-policy cohort E) XGBT in post-policy cohort F) XGBT in pre-policy cohort.

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Although we begin with the caveat that the variables assessed by these survival machine learning models should not be taken as causal nor be interpreted too heavily, we briefly highlight two observations. The first is that ECMO was considered an important variable by RSF in the pre-policy cohort, but not in the post-policy cohort. Given the ongoing debate on whether ECMO is associated with worse post-transplant outcomes [32,34–39], the absence of ECMO in the contemporary post-policy model may suggest that it is no longer a highly

discriminative variable of one-year mortality relative to other variables. Whether this is due to improvements in ECMO management or increased relative usage of IABP remains a future direction of research. Additionally, HIV status was no longer considered an important variable in RSF and Cox Boost in the post-policy cohort. Although one study looked found no differences in transplantation outcomes for HIV-positive and -negative patients overall since 1987 [40], a question arises as to whether contemporary management of HIV patients has improved.

Study limitations

First, as with most national registries, SRTR is susceptible to data entry errors and missing data regarding organ recovery or failure necessary to create continuously updated mortality estimates. However, the SRTR conducts edit checks, validation of data at time of entry, and internal verification when there are outliers. Second, the data did not include known prognostic risk factors such as natriuretic peptides, serum sodium, and specific hemodynamic data such as heart rate, although these are usually available and collected at individual centers. In addition, SRTR does not include other variables such as patient adherence and granular clinical decision-making variables that likely affect survival. Notably, status justification form variables had a high proportion of missingness (above 90%), so they were not included in the predictor set. We would also like to mention that we did not have data that adequately capture social determinants of health relating to access to care or socioeconomic status. Third, this study is exploratory, and the intent was not to build a prediction model for individual prediction of mortality risk but rather to identify potentially important variables associated with mortality. Fourth, machine learning models are often more difficult to interpret than linear models such as Cox since many hyperparameters can be tuned and there is no easily interpretable measure of directional association. Fifth, we assumed that there was only missingness at random and non-informative censoring. However, there may be informative censoring, since early post-transplant deaths are more likely to be recorded than routine follow-up appointments in the first year after transplantation. Sixth, we did not have access to an external heart transplant dataset, which is needed to generalize our findings to populations outside of the United States. Future studies can test survival machine learning methods on prospectively collected SRTR data or more carefully assess how associations of individual predictive variables have changed following the UNOS 2018 policy change.

Conclusions

In this benchmark prognostic study, we demonstrated that machine learning models demonstrate reasonable one-year mortality prediction and can help reveal complex relationships between predictor variables and mortality. We show differences in important variables such as ECMO and HIV status between the post-policy and pre-policy eras, which showcases the ability of machine learning models to generate future hypotheses for research and suggest the need for continual updating of existing models. As clinical care practices continue to evolve, exploring computational time-to-event algorithms to develop mortality risk prediction models with more accumulated data is warranted.

Supporting information

S1 File.
(DOCX)

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